LAB-3

Modified ε-Greedy Strategy for Multi-Armed Bandit.

Calculations.

Given, $M=3$, $N_{play}=10$, $\epsilon=0.2$

Machines: A, B, C

The incremental Sample-average formula is;

$$Q_{new} = Q_{old} + \frac{R - Q_{old}}{N}$$

play-1 - machine A

Random Number : $r = 0.6394$

Since $0.6394 > 0.2$, exploitation is Performed Machine A is selected and gives reward 3.

$N(A) = 1$

$Q(A) = 0 + \frac{3-0}{1} = 3.0$

Play 2 - machine B

Random number : $r = 0.0250$

Since $0.0250 < 0.2$, exploration is Performed. The highest -Q, Machine A is executed and C is selected.

Reward = 4

$N(C) = 1$

$Q(C) = 0 + \frac{6-0}{1}$

$= 6.0$

Play 3 - Machine C

Machine C is selected again and receives award 3.

Before update : $Q(C) = 4$

$N(C) = 1$

After selection : $N(C) = 2$

$\therefore Q(C) = 4 + \frac{3-4}{2}$

$= 4 - 0.5$

$\Rightarrow 3.5$

Final results

Machine	N	R	Q
A	4	14	3.5
B	3	18	6.0
C	3	12	4.0

Total Reward : $3+4+3+4+2+5+6+5+7+5 = 44$

Machine B is the best Performing machine.

$\boxed{Q(B) = 6.0}$